

Data Analytics - Bayesian Modelling of River Flow Anomaly in the Vistula Basin

22.06.2026

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Agenda

1. Problem & Motivation
2. Data Source
3. Basic Model
4. Extended Model
5. Prior Predictive Checks
6. Posterior Analysis & Diagnosis
7. Model Comparison
8. Conclusions & Limitations

1. Problem & Motivation

Motivation:

To build a probabilistic reference model that learns what “normal” river flow looks like at each gauging station, so that hydrological services can automatically flag unusually high flows, detect problematic sensors, and support decisions on where flood-protection infrastructure may require closer assessment.

Our Approach & Objective:

- Estimating station-level log-normal flow distributions with partial pooling through shared river- and network-level hyperparameters.
- Learning what “normal” flow looks like for each station based on 2023 data, then deriving local 95% and 99% anomaly thresholds.
- Testing these station-specific thresholds on a real 2024 hold-out dataset to assess calibration and detect unusually high-flow observations.

Our models build a probabilistic reference baseline for river flows. They estimate what flow values are typical for each gauging station, how variable those flows are, and how unlikely a given observed flow is relative to the station- and river-specific reference distribution.



2. Data Source



Our data was obtained from [Global Runoff Data Centre](#). We requested specific Vistula Basin flow data over the last few decades via email. To account for potential changes caused by modern dams, we focused our analysis on data from 2023.

Dear Jerzyna, Pawel,

Your download has been prepared and will be available in a few minutes.

The file size is 11 MB. Please click this link to download: https://portal.grdc.bafg.de/grdc-download/external/2081d39d-365b-4e34-8f78-54888eced34b/2026-06-03_09-01.zip

Your download will be available for 48 hours.

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GRDC Team

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Disclaimer: While GRDC makes every effort to eliminate errors from the database, there may be errors in the data unknown to GRDC. Neither GRDC nor its sponsors can be held responsible for the consequences of the use of GRDC data.

```
dataset > 6458075_Q_Day.Cmd.txt > data
1 # Title: GRDC STATION DATA FILE
2 # -----
3 # Format: DOS-ASCII
4 # Field delimiter: ;
5 # missing values are indicated by -999.000
6 #
7 # file generation date: 2026-06-03
8 #
9 # GRDC-No.: 6458075
10 # River: BUG
11 # Station: KRYLOW
12 # Country: PL
13 # Latitude (DD): 50.67739
14 # Longitude (DD): 24.06277
15 # Catchment area (km²): 7147.22
16 # Altitude (m ASL): -999.00
17 # Next downstream station: -
18 # Remarks:
19 # Owner of original data: Poland - Institute of Meteorology and Water Management
20 #*****
21 #
22 # Data Set Content: MEAN DAILY DISCHARGE (Q)
23 # -----
24 # Unit of measure: m³/s
25 # Time series: 2006-11 - 2024-10
26 # No. of years: 19
27 # Last update: 2026-05-11
28 #
29 # Table Header:
30 # YYYY-MM-DD - Date
31 # hh:mm - Time
32 # Value - original (provided) data
33 #*****
34 #
35 # Data lines: 6575
36 # DATA
37 YYYY-MM-DD;hh:mm; Value
38 2006-11-01;--:--; 24.600
39 2006-11-02;--:--; 24.600
40 2006-11-03;--:--; 24.000
41 2006-11-04;--:--; 23.800
```

Initially, the raw data required structural preparation as it was provided in a legacy DOS-ASCII format. After cleaning the dataset, which involved removing stations with incomplete records for the 2023–2024 period and filtering out missing values the data was converted into standardized CSV files for subsequent analysis.

```
dataset_cleaned_csv > KRYLOW_BUG.csv > data
1 River,Station,Country,Latitude_DD,Longitude_DD,Catchment_area_km2,Altitude_m_AS,Date,Time,Value
2 BUG,KRYLOW,PL,50.67739,24.06277,7147.22,-999.00,2023-01-01,--:--;65.200
3 BUG,KRYLOW,PL,50.67739,24.06277,7147.22,-999.00,2023-01-02,--:--;67.000
4 BUG,KRYLOW,PL,50.67739,24.06277,7147.22,-999.00,2023-01-03,--:--;68.500
5 BUG,KRYLOW,PL,50.67739,24.06277,7147.22,-999.00,2023-01-04,--:--;70.100
6 BUG,KRYLOW,PL,50.67739,24.06277,7147.22,-999.00,2023-01-05,--:--;71.100
7 BUG,KRYLOW,PL,50.67739,24.06277,7147.22,-999.00,2023-01-06,--:--;70.700
8 BUG,KRYLOW,PL,50.67739,24.06277,7147.22,-999.00,2023-01-07,--:--;69.400
9 BUG,KRYLOW,PL,50.67739,24.06277,7147.22,-999.00,2023-01-08,--:--;68.100
10 BUG,KRYLOW,PL,50.67739,24.06277,7147.22,-999.00,2023-01-09,--:--;66.800
11 BUG,KRYLOW,PL,50.67739,24.06277,7147.22,-999.00,2023-01-10,--:--;65.600
12 BUG,KRYLOW,PL,50.67739,24.06277,7147.22,-999.00,2023-01-11,--:--;64.400
13 BUG,KRYLOW,PL,50.67739,24.06277,7147.22,-999.00,2023-01-12,--:--;62.900
14 BUG,KRYLOW,PL,50.67739,24.06277,7147.22,-999.00,2023-01-13,--:--;61.600
15 BUG,KRYLOW,PL,50.67739,24.06277,7147.22,-999.00,2023-01-14,--:--;59.700
16 BUG,KRYLOW,PL,50.67739,24.06277,7147.22,-999.00,2023-01-15,--:--;57.600
17 BUG,KRYLOW,PL,50.67739,24.06277,7147.22,-999.00,2023-01-16,--:--;56.100
18 BUG,KRYLOW,PL,50.67739,24.06277,7147.22,-999.00,2023-01-17,--:--;55.000
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 52855 entries, 0 to 52860
Data columns (total 10 columns):
#   Column                Non-Null Count  Dtype
---  ---
0   River                  52855 non-null  object
1   Station                52855 non-null  object
2   Country                52855 non-null  object
3   Latitude_DD            52855 non-null  float64
4   Longitude_DD           52855 non-null  float64
5   Catchment_area_km2     52855 non-null  float64
6   Altitude_m_AS         52855 non-null  float64
7   Date                   52855 non-null  datetime64[ns]
8   Time                   52855 non-null  object
9   Value                  52855 non-null  object
dtypes: datetime64[ns](1), float64(4), object(5)
memory usage: 4.4+ MB
None
```

To prepare the data for the Bayesian workflow, we consolidated all processed station records into a single long-format dataset.

3. Basic Model - Baseline Hierarchical Model

Baseline Hierarchical Model - Given the right-skewed nature of hydrology (where most days are characterized by low base flows, and floods represent extreme, rare deviations), a log-normal distribution was utilized. The model operates on a continuous dataset representing daily measurements from hydrological stations, which must consist strictly of positive values ($y > 0$).

A centered parameterization was implemented, providing optimal parameter space geometry in a data-rich regime of over 25k observations. Additionally, the variance scale is modeled in the logarithmic space, which mathematically guarantees that the standard deviation will always be positive upon applying the exponential function.

While this hierarchical log-normal model demonstrates excellent geometric convergence and effective spatial pooling, it operates under the strict assumption of conditional independence among daily observations. In physical hydrology, daily river discharges exhibit strong temporal autocorrelation due to catchment routing processes and groundwater recession.

To ensure methodological integrity, it is crucial to explicitly define the scope of this baseline framework: it is not designed for hydrograph routing or tracking day-to-day temporal flow variations. Its primary function is as a stationary marginal density estimator. By successfully defining the overall climatological probability distribution of flow magnitudes across the Polish gauging network.

TLDR:

This baseline model implements a hierarchical log-normal architecture with centered parameterization to capture the right-skewed spatial distribution of river flows. While it assumes conditional independence and omits temporal autocorrelation, it successfully fulfills its scope as a robust stationary marginal density estimator.

Model Equations:

1. Likelihood:

$$y_n \sim \text{LogNormal}(\mu_{s[n]}, \sigma_{s[n]})$$

2. Station Level (Centered parameterization):

$$\mu_s \sim \mathcal{N}(\mu_{\text{global}}, \tau_\mu)$$

$$\log \sigma_s \sim \mathcal{N}(\log \sigma_{\text{global}}, \tau_\sigma)$$

3. Noise scale transformation:

$$\sigma_s = \exp(\log \sigma_s)$$

- N – Total number of observations in the training partition, i.e. daily measurements from 2023 used for posterior estimation.
- S – Number of unique gauging stations.
- y – Observed river flow [m^3/s].
- station_id – Index vector mapping each observation y_n to a specific station $s \in \{1, \dots, S\}$.

4. Extended Model - Nested Hierarchical Hybrid Model

The extended model introduces an additional grouping layer the individual river level (R), which allows local parameters to be stabilized and enables statistical knowledge transfer within each river system.

The data structure is based on a cascading relationship between observations and stations, and between stations and specific rivers. The input data filtering step removes missing placeholder values and enforces the strictly positive support required by the log-normal likelihood.

TLDR:

This extended architecture implements a three-level hierarchy (Global $\rightarrow$ River $\rightarrow$ Station)

To ensure numerical stability of MCMC sampling and eliminate divergent transitions, the model uses a **hybrid parameterization architecture**, adjusting the type of parameterization to the data density at each hierarchical level:

1. **Higher level (Global $\rightarrow$ River):** This level uses a **non-centered parameterization (NCP)**. Since the number of stations assigned to a single river is relatively small, the hyperprior distribution dominates the likelihood. Factoring out the variance helps remove the geometry of Neal's funnel.
2. **Lower level (River $\rightarrow$ Station $\rightarrow$ Data):** This level uses a **centered parameterization (CP)**. Since each station contains thousands of daily observations ($N_s \gg 300$), the strong likelihood tightly links local parameters to the distribution of the parent river, which supports high effective sample sizes and helps prevent skewed nonlinear correlations.

- N – Number of daily observations in the 2023 training partition used for posterior estimation ($N_{\text{train}} = 25,180$).
- S – Number of unique gauging stations ($S = 69$).
- R – Number of unique rivers grouping the gauging stations.
- y_n – Observed daily river flow for observation n [m^3/s].
- $\text{station}_i\text{d}[n]$ – Vector mapping observation n to a specific station $s \in 1, \dots, S$.
- $\text{river}_i\text{d}[s]$ – Vector mapping station s to its parent river $r \in 1, \dots, R$.

The indexing path is cascading: observation n belongs to station $s[n]$, which in turn belongs to river $r[s]$. This can be written compactly as $r[s[n]]$.

Model equations

1. Likelihood:

$$y_n \sim \text{LogNormal}(\mu_{s[n]}, \sigma_{s[n]})$$

2. Station level — Centered parameterization:

The parameters of each station are sampled directly from the distribution of the parent river. The free variable in the model is $\log \sigma_s$, which guarantees the positivity of σ_s after applying the exponential transformation:

$$\mu_s \sim \mathcal{N}(\mu_{r[s]}, \tau_{\mu, \text{sta}})$$

$$\log \sigma_s \sim \mathcal{N}(\log \sigma_{r[s]}, \tau_{\sigma, \text{sta}})$$

$$\sigma_s = \exp(\log \sigma_s)$$

3. River level — Non-centered parameterization:

River-level parameters are constructed as deterministic shifts from the global reference point, scaled by standardized auxiliary variables z :

$$\mu_r = \mu_{\text{global}} + \tau_{\mu, \text{riv}} \cdot z_{\mu, r}$$

$$\log \sigma_r = \log \sigma_{\text{global}} + \tau_{\sigma, \text{riv}} \cdot z_{\sigma, r}$$

$$\sigma_r = \exp(\log \sigma_r)$$

4. Auxiliary variables in standardized space:

$$z_{\mu, r} \sim \mathcal{N}(0, 1), \quad z_{\sigma, r} \sim \mathcal{N}(0, 1)$$

5. Prior Predictive Checks

Basic Model

Prior Predictive Checks (PPC) were used to calibrate weakly informative priors, balancing physical hydrology with computational stability in Stan. While the Log-Normal distribution mathematically enforces strictly positive river flow ($Q > 0$), hyperparameters were tuned to truncate absurd right-tail values ("prior explosion") and establish a realistic 99th percentile ceiling around 5000 m³/s to safely capture historical flood events

$$y_n \sim \text{LogNormal}(\mu_{s[n]}, \sigma_{s[n]})$$

$$\mu_s \sim \mathcal{N}(\mu_{\text{global}}, \tau_\mu)$$

$$\log \sigma_s \sim \mathcal{N}(\log \sigma_{\text{global}}, \tau_\sigma)$$

$$\sigma_s = \exp(\log \sigma_s)$$

Parameter	Distribution	Physical Justification
μ_{global}	$\mathcal{N}(3, 1.2)$	Global average flow on a logarithmic scale. A standard deviation of 1.2 allows the model to consider both small streams and massive rivers at their mouths.

τ_{μ}
 $\text{Half-}\mathcal{N}(0, 1.2)$

Tolerance for deviations from the global average (inter-station variability). Expanded to capture the vast hydrological diversity of the target network.

 $\log(\sigma_{\text{global}})$
 $\mathcal{N}(0, 0.5)$

Baseline for the standard deviation of measurement noise. A tight standard deviation of 0.5 prevents the right tail of the log-normal distribution from exploding.

 τ_{σ}
 $\text{Half-}\mathcal{N}(0, 0.5)$

Variability in water dynamics dispersion (variance) across individual stations.

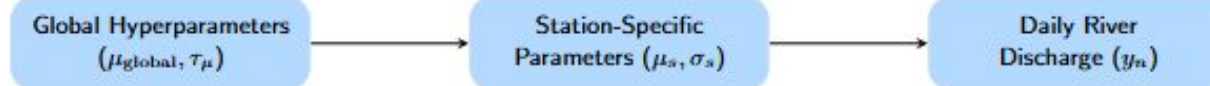
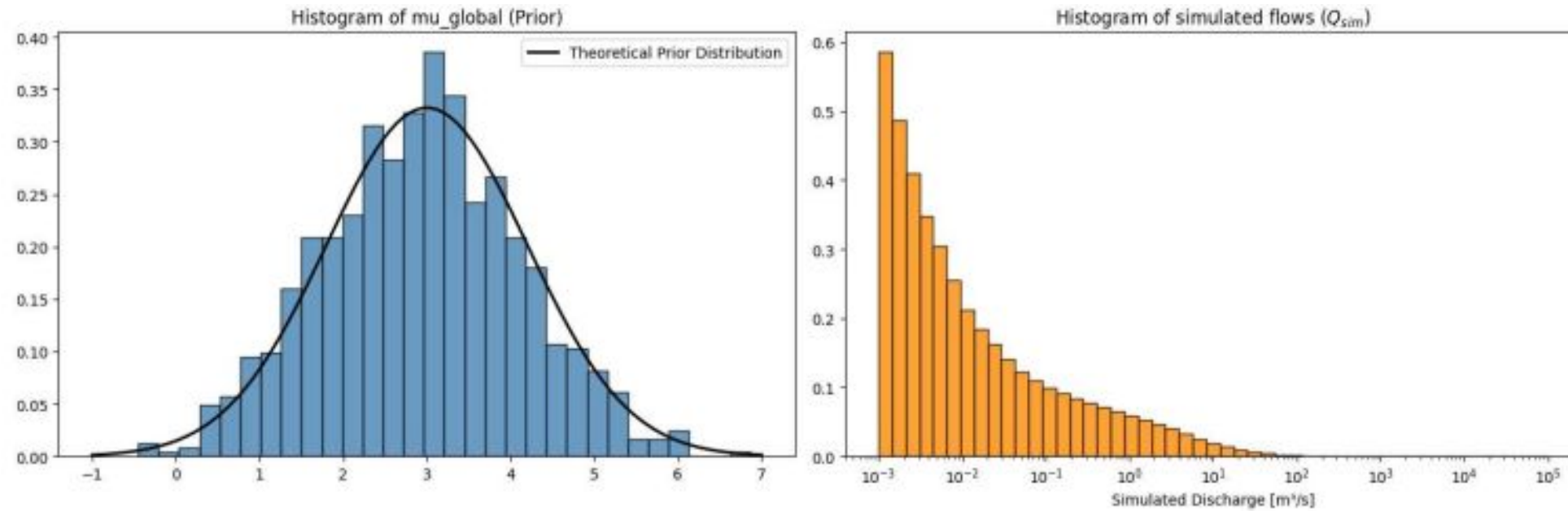


Figure 1: Simplified conceptual DAG of the Baseline Hierarchical Log-Normal Model.



```
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PRIOR PREDICTIVE CHECK SUMMARY STATISTICS:
  Median simulated discharge:          18.73 m3/s
  95th percentile (high-flow regime): 652.91 m3/s
  99th percentile (extreme events):   5150.96 m3/s
=====
```

Extended Model

$$y_n \sim \text{LogNormal!}(\mu_{s[n]}, \sigma_{s[n]})$$

$$\mu_s \sim \mathcal{N}(\mu_{r[s]}, \tau_{\mu, \text{sta}})$$

$$\log \sigma_s \sim \mathcal{N}(\log \sigma_{r[s]}, \tau_{\sigma, \text{sta}})$$

$$\sigma_s = \exp(\log \sigma_s)$$

$$\mu_r = \mu_{\text{global}} + \tau_{\mu, \text{riv}} \cdot z_{\mu, r}$$

$$\log \sigma_r = \log \sigma_{\text{global}} + \tau_{\sigma, \text{riv}} \cdot z_{\sigma, r}$$

$$\sigma_r = \exp(\log \sigma_r)$$

$$z_{\mu, r} \sim \mathcal{N}(0, 1), \quad z_{\sigma, r} \sim \mathcal{N}(0, 1)$$

The global-level hyperparameter distributions define the central reference point for the entire country. The τ parameters at each level control the strength of shrinkage: the smaller the value of τ , the more strongly stations are pulled toward the mean of their parent river.

Parameter	Distribution	Range and role in the hierarchy
μ_{global}	$\mathcal{N}(3, 1.2)$	Global reference point for the logarithm of river flows at the national scale.
$\log \sigma_{\text{global}}$	$\mathcal{N}(0, 0.5)$	Central baseline value for the noise scale, i.e. within-station variance.

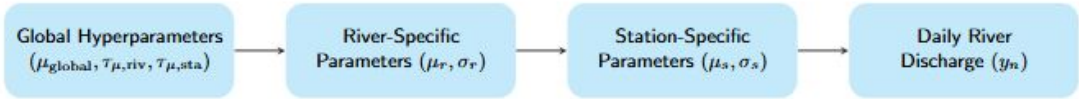
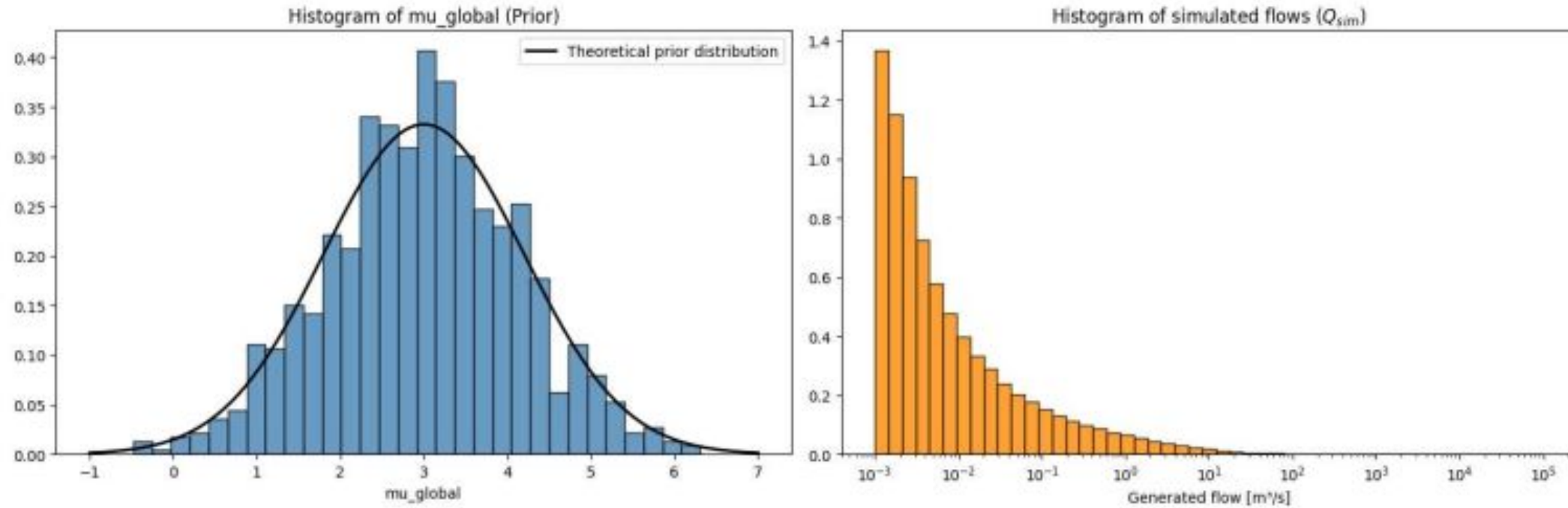


Figure 2: Simplified conceptual DAG of the Extended Nested Hierarchical Model.

$\tau_{\mu,\text{riv}}$	Half- $\mathcal{N}(0, 1.2)$	Scale of variability in mean flows between rivers.
$\tau_{\sigma,\text{riv}}$	Half- $\mathcal{N}(0, 0.5)$	Scale of variability in flow dispersion dynamics between rivers.
$\tau_{\mu,\text{sta}}$	Half- $\mathcal{N}(0, 1.2)$	Local variability of station means within the same river.
$z_{\mu,r}$	$\mathcal{N}(0, 1)$	Standardized location deviation for river r .
$z_{\sigma,r}$	$\mathcal{N}(0, 1)$	Standardized scale deviation for river r .



```
=====
PRIOR PREDICTIVE CHECK REPORT (extended model):
  • Median of generated flows: 19.99 m3/s
  • 95th percentile of flow in PPC: 1263.26 m3/s
  • 99th percentile (extreme): 12330.29 m3/s
=====
```

Prior Predictive Check Analysis

Basic Model

Our priors were weakly informative, but:

$\mu_{\text{global}} \sim \mathcal{N}(3, 1.2)$ — covers full range from mountain streams to major rivers on log scale

$\tau_{\mu}, \tau_{\sigma} \sim \text{Half-}\mathcal{N}(0, 1.2 / 0.5)$ — controls inter-station pooling strength

$\log \sigma_{\text{global}} \sim \mathcal{N}(0, 0.5)$ — prevents right-tail explosion of the log-normal

Extended Model

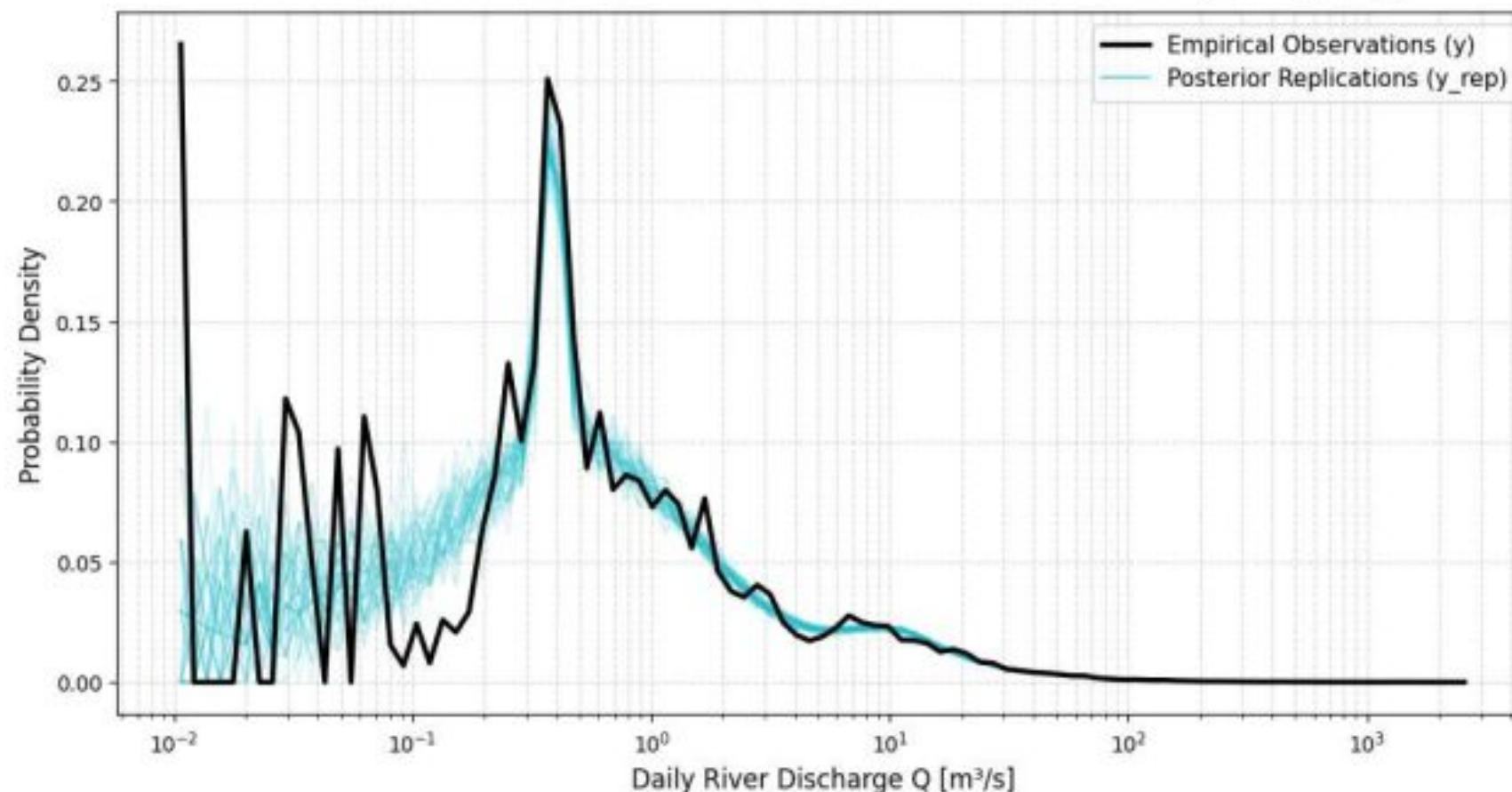
Parameter	Prior	Posterior mean	94% HDI
μ_{global}	$\mathcal{N}(3, 1.2)$	2.285	[1.72, 2.85]
$\log \sigma_{\text{global}}$	$\mathcal{N}(0, 0.5)$	-0.454	[-0.60, -0.30]
$\tau_{\mu, \text{riv}}$	$\text{Half-}\mathcal{N}(0, 1.2)$	1.212	[0.74, 1.68]
$\tau_{\sigma, \text{riv}}$	$\text{Half-}\mathcal{N}(0, 0.5)$	0.447	[0.33, 0.57]
$\tau_{\mu, \text{sta}}$	$\text{Half-}\mathcal{N}(0, 1.2)$	1.453	[1.17, 1.75]
$\tau_{\sigma, \text{sta}}$	$\text{Half-}\mathcal{N}(0, 0.5)$	0.178	[0.13, 0.22]

$\tau_{\mu, \text{sta}} > \tau_{\mu, \text{riv}}$ — station placement explains more location variance than river identity

$\tau_{\sigma, \text{riv}} > \tau_{\sigma, \text{sta}}$ — flow volatility is a *river-basin* property, not station-specific

6. Posterior Analysis & Diagnosis

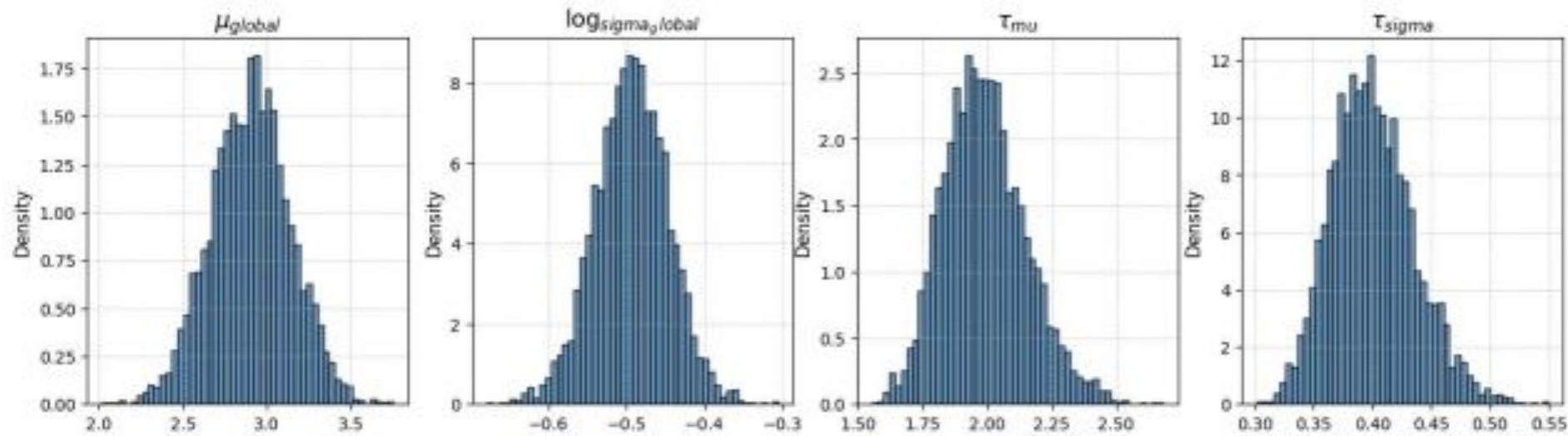
Posterior Predictive Check - Hierarchical Baseline Model (Histogram)



The baseline model demonstrates valid predictive performance across the primary and high-flow regimes, accurately capturing the central hydrological mode and tightly enveloping the heavy right tail to model extreme flood risks.

However, it systematically misfits the very low flow spectrum ($Q \leq 0.1 \text{ m}^3/\text{s}$), where the continuous log-normal distribution blurs and overestimates density.

Marginal Posterior Distributions - Baseline Model

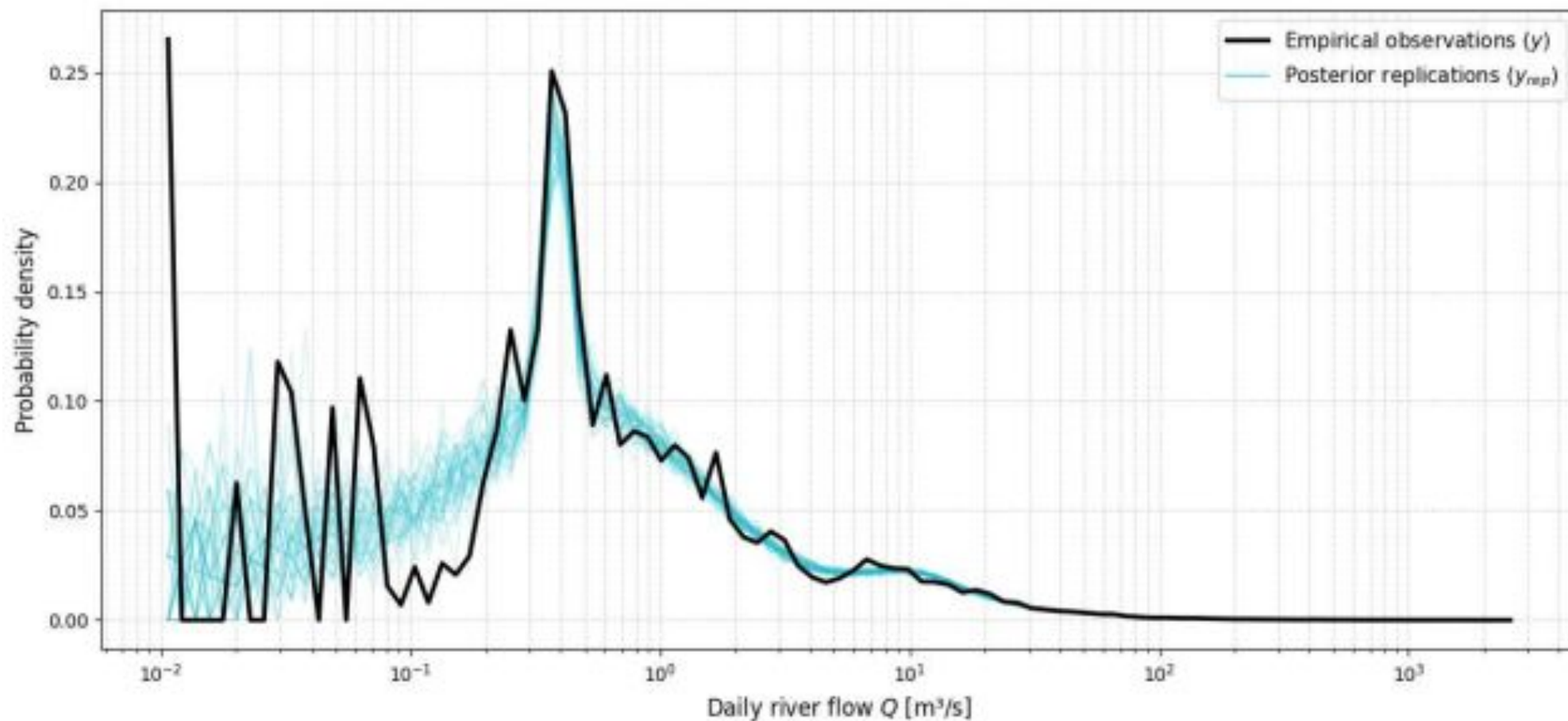


wide distribution = greater parameter uncertainty

narrow distribution = parameter well defined by the data

distribution position = typical/average parameter value

Posterior Predictive Check - Extended Model (Nested Hierarchical)



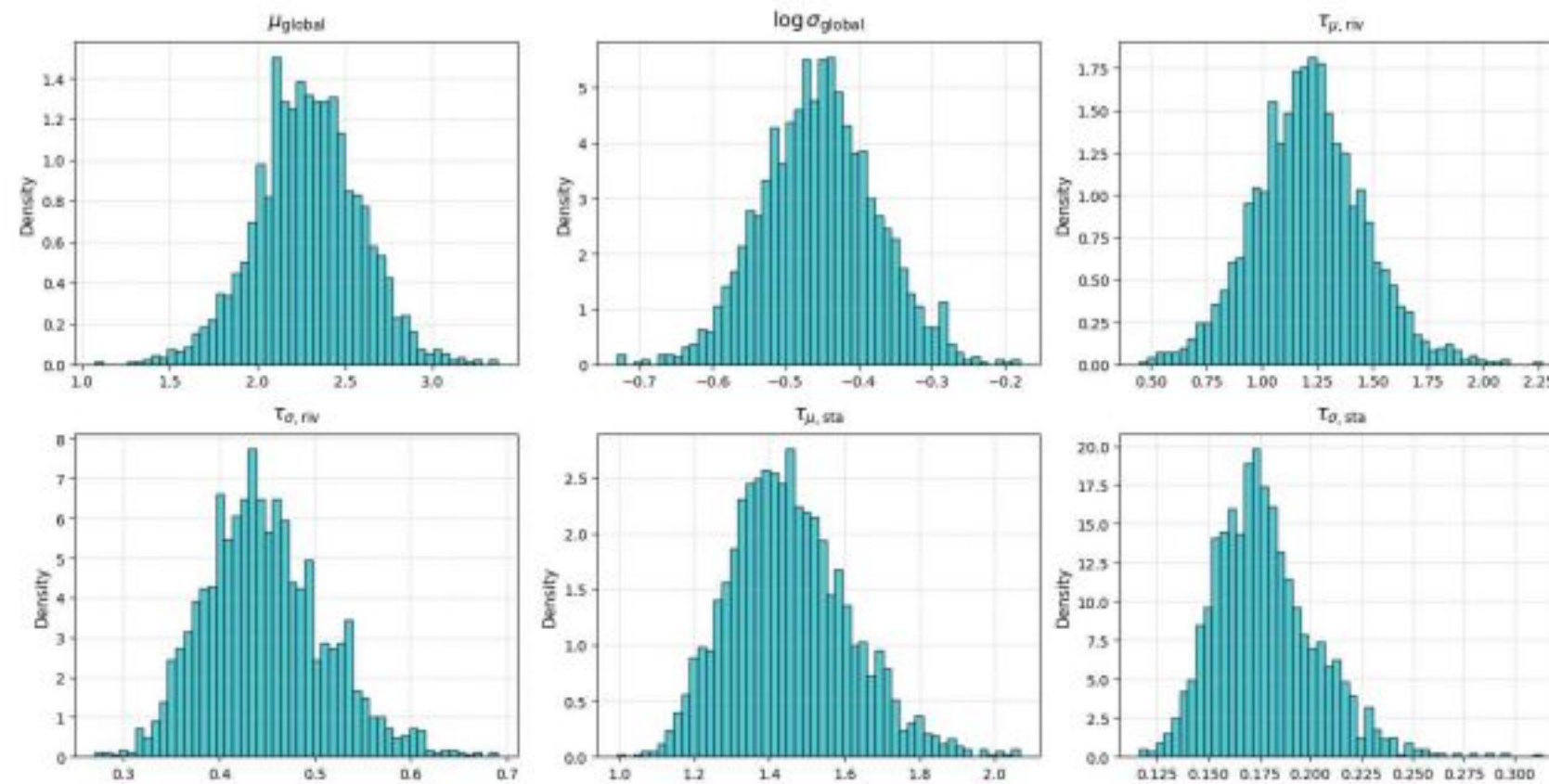
The extended hierarchy lowers the global median flow estimate to 9.8 m³/s by separating station- and river-level variation.

Key hydrological insight:

- typical flow is mostly station-location dependent
- volatility is mostly shared within the river basin

The model captures the main flow regime and heavy right tail, but the low-flow artefact below 0.1 m³/s remains a limitation.

Marginal Posterior Distributions - Extended Model (Nested Hierarchical)



The global typical flow corresponds to a median of about 10 m³/s, while the global log-scale variability is about 0.64.

The most important insight comes from the hierarchy parameters:

- **$\tau_{\mu, \text{station}} > \tau_{\mu, \text{river}}$**
This means that the typical flow level depends strongly on the exact gauge location.
- **$\tau_{\sigma, \text{river}} > \tau_{\sigma, \text{station}}$**
This means that flow volatility is more a property of the river or basin than of a single station.

So, in simple terms: where the gauge is located matters most for the typical flow level, but the variability of flow is mostly shared within the river system.

Model validation on 2024 data

GLOBAL FLOOD THRESHOLD VALIDATION REPORT (2024):

- Total number of analyzed station-days (N): 22203
- Detected anomalies (flow > 95% threshold): 2006 days (9.03% of total)
- Detected flood extremes (99% threshold): 748 days (3.37% of total)

Detailed validation report for 69 stations saved to: 'anomaly_validation_report_2024.csv'

TOP 10 STATIONS WITH THE HIGHEST NUMBER OF ANOMALIES IN 2024:

	Station	Measurement_Days	Anomalies_95_Sum	Percentage_Anomalies_95%
0	OJCOW	305	288	94.426230
1	PTAKI	366	120	32.786885
2	DOBRYLAS	305	116	38.032787
3	TUCHOLA	366	95	25.956284
4	BRODNICA	366	64	17.486339
5	ELGISZEWO	305	57	18.688525
6	BORKOWO	305	56	18.360656
7	LOCHOW	366	50	13.661202
8	KRASNYSTAW	366	49	13.387978
9	JELEN	305	46	15.081967

GLOBAL FLOOD THRESHOLD VALIDATION REPORT - EXTENDED MODEL (2024):

- Total number of analyzed station-days (N): 22203
- Detected anomalies (flow > 95% threshold): 1994 days (8.98% of total)
- Detected flood extremes (99% threshold): 745 days (3.36% of total)

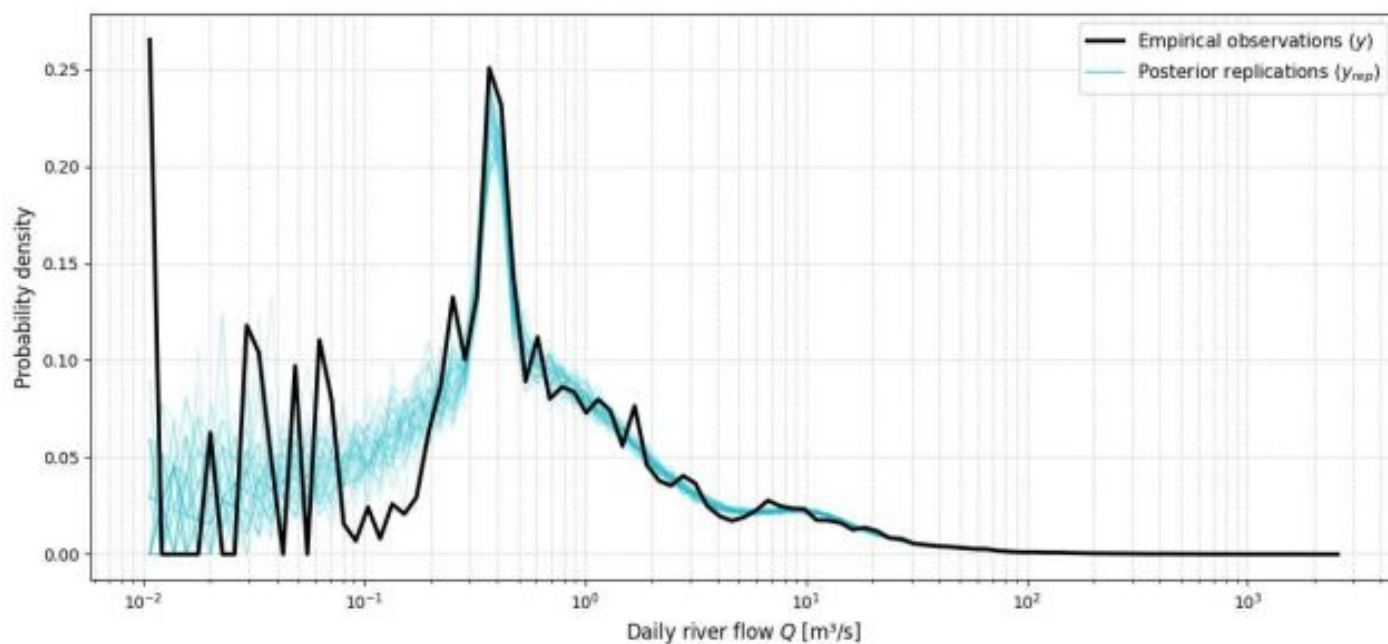
Detailed validation report for 69 stations saved to: 'anomaly_validation_report_extended_2024.csv'

TOP 10 STATIONS WITH THE HIGHEST NUMBER OF ANOMALIES IN 2024 (EXTENDED MODEL):

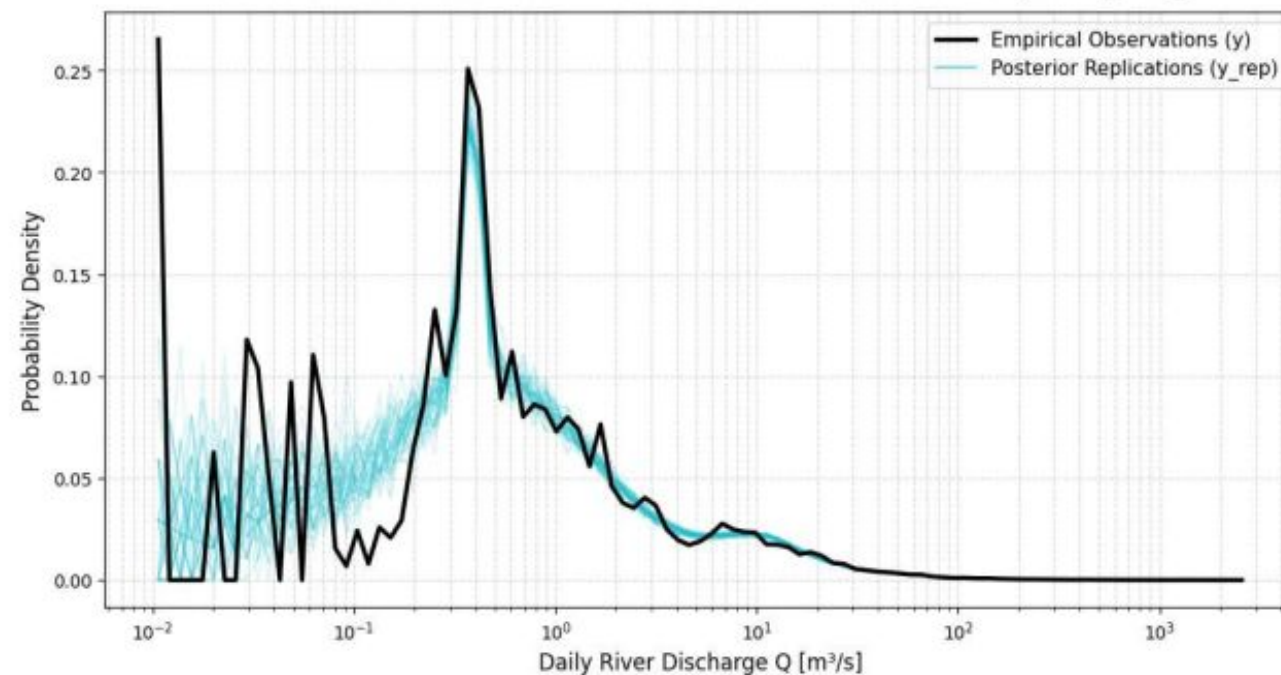
	Station	Measurement_Days	Anomalies_95_Sum	Percentage_Anomalies_95%
0	OJCOW	305	288	94.426230
1	PTAKI	366	120	32.786885
2	DOBRYLAS	305	116	38.032787
3	TUCHOLA	366	95	25.956284
4	BRODNICA	366	63	17.213115
5	ELGISZEWO	305	57	18.688525
6	BORKOWO	305	56	18.360656
7	LOCHOW	366	49	13.387978
8	KRASNYSTAW	366	49	13.387978
9	JELEN	305	46	15.081967

7. Model Comparison

Posterior Predictive Check - Extended Model (Nested Hierarchical)



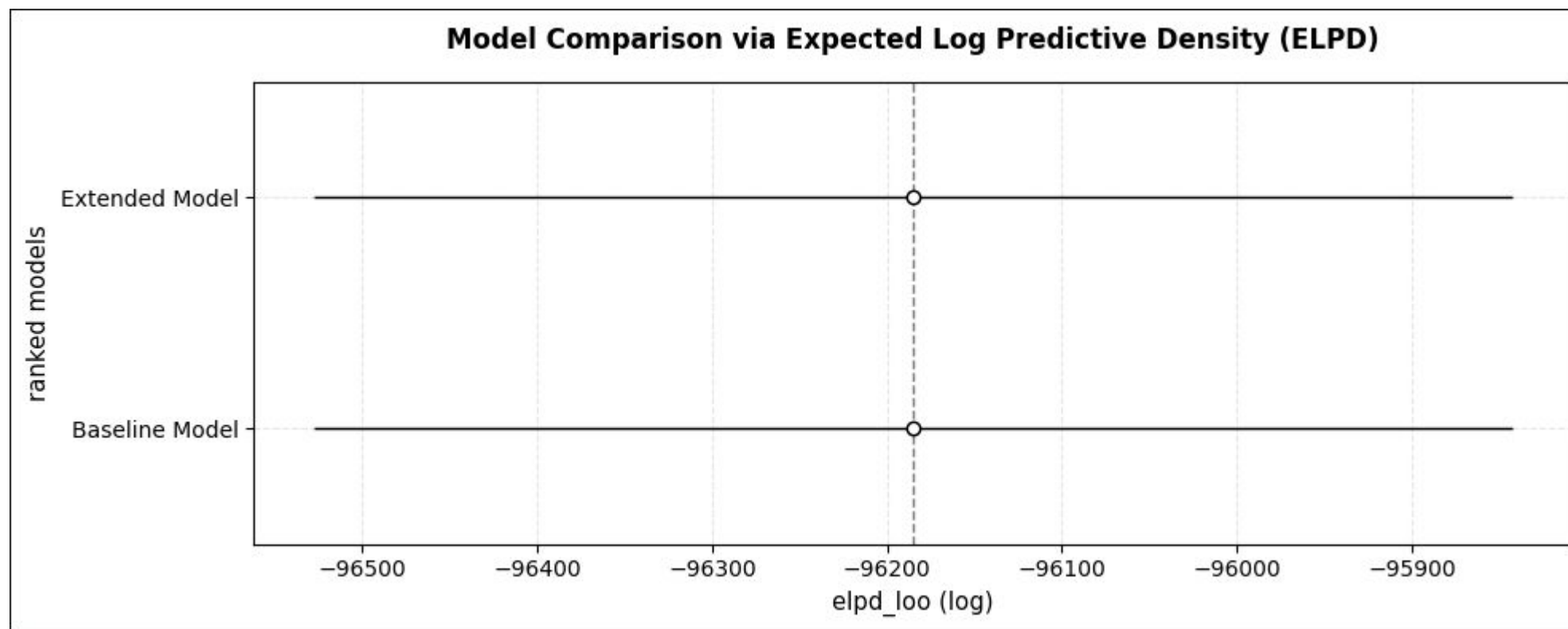
Posterior Predictive Check - Hierarchical Baseline Model (Histogram)



Global Hyperparameter Comparison

Hyperparameter	Baseline Model (Global → Station)	Extended Model (Global → River → Station)	Hydrological & Structural Insight			
μ_{global} (Mean)	2.901 (HDI: [2.473, 3.355]) ≈ 18.19 m ³ /s	2.285 (HDI: [1.716, 2.845]) ≈ 9.82 m ³ /s	Downward Shift: Nesting stations within parent rivers prevents clusters of highly active or closely located stations from disproportionately driving the national baseline median upward.	τ_{μ} (Location Variance)	1.990 (HDI: [1.700, 2.297])	Partitioned: $\tau_{\mu,\text{riv}} = 1.212$ $\tau_{\mu,\text{sta}} = 1.453$
$\log \sigma_{\text{global}}$ (Noise)	-0.494 (HDI: [-0.586, -0.407]) ≈ 0.61	-0.454 (HDI: [-0.596, -0.298]) ≈ 0.64	Highly Stable: The global baseline for within-station environmental noise and measurement variance remains robust against structural architectural changes.	τ_{σ} (Scale Variance)	0.400 (HDI: [0.334, 0.465])	Partitioned: $\tau_{\sigma,\text{riv}} = 0.447$ $\tau_{\sigma,\text{sta}} = 0.178$
				Decomposition: Total spatial variance remains consistent ($\sqrt{1.212^2 + 1.453^2} \approx 1.89$), but the extended model reveals that localized station geomorphology ($\tau_{\mu,\text{sta}}$) contributes more to flow variance than macro-river basin differences ($\tau_{\mu,\text{riv}}$).		
				Dynamic Realignment: The variation in flow volatility is predominantly dictated by river-basin characteristics ($\tau_{\sigma,\text{riv}}$) rather than differences between gauges on the same river ($\tau_{\sigma,\text{sta}}$).		

Model Configuration	Empirical 95% Interval Coverage	Deviation from Target	Calibration Quality
Theoretical Target	95.00%	0.00 pp	Perfect Calibration
Extended Model	89.98%	-5.02 pp	Moderate Undercoverage
Baseline Model	89.95%	-5.05 pp	Moderate Undercoverage



These uncertainty intervals are relatively wide because the predictive density is estimated over a large and heterogeneous hydrological dataset. The extended model is ranked first by PSIS-LOO, but the estimated difference between the two models is very small. The Δ_{ELPD} is approximately 0.17, while the standard error of the difference is approximately 0.91. Because the uncertainty of the difference is larger than the difference itself, the comparison does not provide strong statistical evidence that the extended model has substantially better predictive performance than the baseline model.

Additionally, the ArviZ output reports Pareto-k warnings for both models. This indicates that some observations are highly influential and that the PSIS-LOO approximation may be unstable for those data points. As a result it should be treated as a useful diagnostic rather than as definitive model-selection proof.

8. Conclusions & Limitations

The 2024 validation results show that the extended model performs very similarly to the baseline model in aggregate threshold exceedance rates. The extended hierarchy slightly reduces the number of validation exceedances, but the improvement is marginal and should not be described as a decisive predictive gain. The main justification for the extended model is therefore methodological: it introduces a more interpretable nested structure by separating station-level and river-level variability. This allows partial pooling within river systems and provides a more hydrologically meaningful representation of the monitoring network. However, the empirical validation results indicate that this additional structure does not substantially improve nationwide threshold calibration in the 2024 hold-out period.

Both models achieve comparable out-of-sample performance: the extended nested hierarchy ranks marginally higher under PSIS-LOO ($\Delta\text{ELPD} = 0.17$), confirming that the added river layer is warranted by structural insight rather than measurable predictive advantage. Pareto-k diagnostics flagged several highly influential observations in both models, indicating that a small number of extreme flow events drive disproportionate uncertainty in the LOO approximation, so results should therefore be interpreted as indicative rather than definitive.

Thank you!

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